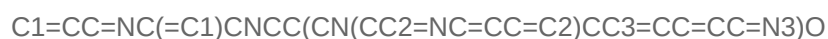
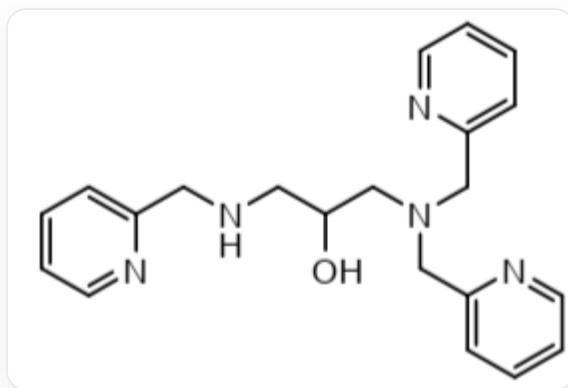


Question

The multidentate ligand **HTPPNOL** (as shown) forms complexes with copper that can be used to simulate the active centers of binuclear copper-containing enzymes.



In all the formed complexes, the rings consisting solely of Cu^{2+} and atoms from **HTPPNOL** are all five-membered rings.

In a sodium acetate solution, **HTPPNOL** reacts with Cu^{2+} to form the binuclear complex $[\text{Cu}_2(\text{TPPNOL})(\text{CH}_3\text{COO})]^{2+}$. In this complex ion, Cu^{2+} exhibits two coordination numbers, one of which is five-coordinate (trigonal bipyramidal geometry), and the complex structure includes four six-membered ring.

Based on the above information, select the correct option.

- A. Another coordination number of Cu^{2+} is 5.
- B. The complex has 5 five-membered rings.
- C. In this complex, the two Cu^{2+} ions have different oxygen coordination numbers.

- D.** When the salt of this complex ion is dissolved in water, only one binuclear complexes with identical coordination numbers for two Cu^{2+} exist in the solution. Moreover, these two complexes can interconvert by adjusting the pH of the solution. In both complexes, the coordination number of Cu^{2+} is 4.
- E.** In the alkaline solution of the Cu^{2+} -HTPPNOL system without other ligands, there exist binuclear complex ions where the coordination number of Cu^{2+} is 5, and the oxygen coordination numbers of the two Cu^{2+} ions in this complex are identical.
- F.** To measure the crystal structure of the complex ion in option E, you can choose a solution with smaller anions to reduce the solubility of the complex ion in the solution and obtain crystals.
- G.** None of the above options is correct.

Answer

Correct Answer: **B**

Detailed Explanation

A new six-membered ring exists in the complex. Based on the information that all rings composed only of Cu^{2+} and atoms from **HTPPNOL** are five-membered rings, this six-membered ring cannot be composed of **HTPPNOL** and Cu^{2+} . Therefore, it must be formed by bridging acetate ions, namely the $\text{Cu} - \text{O} - \text{C} - \text{O} - \text{Cu}$ bridge.

CHECKPOINT

1 PTS

The six-membered ring is formed by bridging acetate ions, namely the $\text{Cu} - \text{O} - \text{C} - \text{O} - \text{Cu}$ bridge.

According to the structure of **HTPPNOL** and the clue that all rings formed by Cu^{2+} and atoms from **HTPPNOL** are five-membered rings, it can be known that N/N or N/O every two carbons in **HTPPNOL** can form a five-membered ring with Cu^{2+} . A total of 5 five-membered rings can be generated.

CHECKPOINT

1 PTS

N/N or N/O every two carbons in **HTPPNOL** can form a five-membered ring with Cu^{2+} , and a total of 5 five-membered rings can be generated.

Based on the distance between Cu^{2+} and the coordinating atoms on **HTPPNOL**, the approximate structure of the dinuclear complex formed by **HTPPNOL** and Cu^{2+} can be inferred. For five-coordinated Cu^{2+} , it is coordinated with two pyridine nitrogens, one amine nitrogen connected to the two methylpyridines, one hydroxyl oxygen in **HTPPNOL**, and one carboxyl oxygen in CH_3COO^- .

CHECKPOINT

2 PTS

The five-coordinated Cu^{2+} is coordinated with two pyridine nitrogens, one amine nitrogen connected to the two methylpyridines, one hydroxyl oxygen in **HTPPNOL**, and one carboxyl oxygen in CH_3COO^- .

The other Cu^{2+} is coordinated with the remaining one pyridine nitrogen, one amine nitrogen connected to the methylpyridine, one hydroxyl oxygen in **HTPPNOL**, and one carboxyl oxygen in CH_3COO^- . Its coordination number is 4.

CHECKPOINT

2 PTS

The other Cu^{2+} is coordinated with one pyridine nitrogen, one amine nitrogen connected to the methylpyridine, one hydroxyl oxygen in **HTPPNOL**, and one carboxyl oxygen in CH_3COO^- , and the coordination number is 4.

Option A: The coordination number of the other Cu^{2+} is 4, A is incorrect.

Option B: The complex has 5 five-membered rings, B is correct.

Option C: The O coordination numbers of Cu^{2+} in the complex are the same, both are 2, C is incorrect.

Option D: Dissolving the salt of the complex ion in water yields two dinuclear complexes with the same Cu^{2+} coordination number. It is possible that H_2O coordinates with Cu^{2+} with a coordination number of 4, resulting in complexes with Cu^{2+} coordination numbers of 5.

CHECKPOINT

1 PTS

H_2O coordinates with Cu^{2+} with a coordination number of 4, and the Cu^{2+} coordination numbers of the complex are all 5.

Among them, the protonation/deprotonation of $\text{H}_2\text{O}/\text{OH}^-$ can be achieved by adjusting the pH, forming these two dinuclear complexes.

CHECKPOINT

1 PTS

The protonation/deprotonation of $\text{H}_2\text{O}/\text{OH}^-$ can be achieved by adjusting the pH to complete the conversion of the two complexes

D is incorrect.

Option E:

In an alkaline solution without other ligands, only OH^- coordinates with Cu^{2+} besides **HPPNOL**.

CHECKPOINT

1 PTS

OH^- coordinates with Cu^{2+} .

It is known that a dinuclear complex with Cu^{2+} coordination numbers of 5 is formed in the system, indicating that one OH^- replaces the position of CH_3COO^- in the original complex and coordinates with two Cu^{2+} respectively.

CHECKPOINT

1 PTS

One OH^- coordinates with two Cu^{2+} respectively.

Another OH^- coordinates with the original Cu^{2+} with a coordination number of 4 to form the complex.

CHECKPOINT

1 PTS

Another OH^- coordinates with the original Cu^{2+} with a coordination number of 4

Adding the hydroxyl oxygen in **HTPPNOL** that coordinates with the two Cu^{2+} , the oxygen coordination numbers of Cu^{2+} in the complex are 2 and 3 respectively.

CHECKPOINT

1 PTS

The oxygen coordination numbers of Cu^{2+} in the complex are 2 and 3 respectively.

E is incorrect.

Option F: According to the Bartolho rule, crystals composed of large anions and large cations have lower solubility. A solution with larger volume anions should be selected to reduce the solubility of the complex ion in the solution to obtain crystals.

CHECKPOINT

1 PTS

According to the Bartolho rule, a solution with larger volume anions should be selected

F is incorrect.

Option G: G is incorrect.